

Phenotyping in the OMOP Common Data Model

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Preface

Is this book for me?

This book is part of the collection of **Tidy R programming in OMOP** see accompanying books:

- (1) Tidy R programming with the OMOP common data model
- (2) Tidy R packages with the OMOP common data model
- (3) Drug Utilisation Studies in the OMOP Common Data Model
- (4) Phenotyping in the OMOP Common Data Model

We've written this book for anyone interested to conduct studies using the OMOP Common Data Model and working with databases using a tidyverse style approach. That is, human centered, consistent, composable, and inclusive (see <https://design.tidyverse.org/unifying.html> for more details on these principles).

New to R? We recommend you compliment the book with R for data science

New to databases? We recommend you take a look at some web tutorials on SQL, such as SQLBolt or SQLZoo

New to the OMOP CDM? We'd recommend you pare this book with The Book of OHDSI

New to the Tidy R in OMOP ecosystem? Probably you should start with the book Tidy R programming with the OMOP common data model before reading this one.

How is the book organised?

Package	Version	Link
CodelistGenerator	4.0.2	
PhenotypeR	0.5.0	
duckdb	1.5.2	
omock	0.6.2	

How is the book organised?

TO ADD

Citation

TO ADD

License

Code

The source code for the book can be found at this Github repository

Packages version

This book is rendered using the following version of packages:

```
Finding R package dependencies ... Done!
```

Note we only included the packages called explicitly in the book.

1. Proposed structure for the book

- Preface
- Introduction
 - What is Phenotyping
 - The phenotyping process
 - Clinical description
- OMOP CDM vocabularies
 - Introduction
 - Download vocabularies
 - Condition based concepts
 - Drug based concepts
 - Measurement based concepts
 - Observation based concepts
 - Procedures based concepts
 - Devices based concepts
- Codes in OMOP CDM
 - Introduction
 - Codelists
 - Codelist with details
 - Concept set expression
 - get candidate codelists
 - Create a condition based codelist (example stroke)
 - Create a
 - Create a drug based codelist (example chemotherapy)

1. Proposed structure for the book

- Create a drug based codelist (sth that we can stratify by brand, route and dose unit)
- Create cohorts
 - Base cohorts
 - Demographic cohorts
 - Inclusion criteria
 - Set the end date
 - Combine cohorts
 - Example cohorts (obesity + causal inference centered on a complicated end date e.g. venla study)
- Phenotype validation
 - Database Feasibility
 - * Introduction
 - * Metadata
 - * The person table
 - * The observation period
 - * Clinical tables
 - * Record trends
 - Codelist diagnostics
 - * Code use
 - * Orphan codes
 - * Drug exposure diagnostics
 - * Measurement diagnostics
 - Cohort diagnostics
 - * Counts and attrition
 - * Characterisation and large scale characterisation
 - * Matched comparison
 - * Overlap and timing
 - * Survival
 - Population diagnostics

1. Proposed structure for the book

- * Incidence
- * Prevalence
- Study workflow and final remarks

```
library(CodelistGenerator)
library(PhenotypeR)
library(omock)
library(duckdb)
```

Loading required package: DBI

Part I.

Introduction

2. Phenotyping process

Part II.

The OMOP CDM vocabularies

3. Download vocabularies

In this chapter we explain how to download the OMOP CDM vocabularies and create an OMOP CDM instance with just the vocabulary tables. We will use this OMOP CDM instance to explore the vocabularies in the next chapters.

The OMOP CDM vocabularies are distributed through Athena this is a platform where you can explore the different vocabularies, perform searches and explore the hierarchy of the vocabularies. If you want to read more about Athena you can read the chapter ... from *The book of OHDSI*.

3.1. Download vocabularies

- Create an account in Athena
- Prepare the vocabularies that you are interested.
- Download
- Extract CPT4 *note this step is only needed if*
- create the cdm_reference

Final remarks

Drug Utilisation Studies in the OMOP Common Data Model aims to ...

Learning more

If you find this book useful then joining the **Tidy R in OMOP** OHDSI working group will likely be of interest. Building on many of the concepts and tools seen in this book, the Tidy R in OMOP OHDSI working group aims to (1) create and promote a unified set of resources to guide Tidy R development in OMOP and support the OHDSI community, and (2) establish an overview of available packages relevant to Tidy R programming in OMOP (tidyverse style packages). If you are interested in joining the working group then please email Martí Català or Raivo Kolde.

Support us

We encourage you to support this work by either citing the book in your papers or documentation, recommending it to your colleagues, or starring the GitHub repository, or simply letting us know how it helped you. Most importantly, please **use it** in research that results in patient benefit.